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Geothermal energy in Turkey and around the World: A review of the literature and an analysis based on Turkey's Vision 2023 energy targets

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ABSTRACT

Turkey's geothermal energy potential suitable for electricity generation was estimated at nearly 4500 MW. In Turkey, work on power production from geothermal energy started in the second half of the 20th century. However, capacity build up and investments accelerated after 2000s. This is mostly related to Turkey's staggering economic growth in the last 10 years, which concurrently increased the country's energy demand. Also, renewable energy, especially geothermal energy, legislations passed and laws enacted in the last 10 years eased the process. The subsidies and supports to the private sector also positively affected the development of geothermal power plants. The growth of Turkey's energy market is a part of Turkey's ambitious Vision 2023 energy targets. One of the main aims of which is to produce 30% of Turkey's electricity demand in 2023 from renewable energy sources. However, geothermal energy has the smallest share (in terms of installed capacity) in the Vision 2023 energy targets set at 600 MW by the year 2023 (target, as of 2015, can be modified). This means geothermal power plants could only provide 0.5% of Turkey's installed capacity, which was estimated as 120,000 MW in the year 2023. In this study, it was pinpointed that geothermal energy installed capacity of Turkey reached to 623.9 MW at the end of 2015. This is critically important for the realisation of the Vision 2023 energy targets, and it is the first renewable energy target that has been fulfilled before 2023. Also, in this study, it was suggested to modify the Vision 2023 targets to increase the geothermal energy installed capacity to 1000 MW. By this way, the share of geothermal energy in Turkey's installed capacity of 2023 could rise to nearly 0.8%.

1. Introduction

Turkey is a developed country and a regional power [1]. Turkey's gross domestic product (GDP) in purchasing power parity (PPP) was estimated at 1.576 trillion US$ in 2015 [2]. Data about Turkey's GDP (PPP) change between 1990 and 2014 is shown in Fig. 1 [3]. Historically, economic growth and energy consumption have been highly correlated [4,5]. There are various number of studies that have examined the relationship between energy consumption and GDP growth [6]. However, conflicting results emerged from these studies both on the existence and the direction of causality [7]. Although there are plenty of studies in the literature that investigated the GDP growth and energy consumption relationship, there are only few studies, which examined the relationship between renewable energy consumption and GDP growth [8]. There are three main types of causal relationship between energy consumption and economic growth in the literature: unidirectional causality, bidirectional causality, and no causality [9]. For policy makers the direction of the causal relationship is imperative [10].

During the last 25 years, Turkey's energy consumption increased parallel to its economic growth. Net electricity consumption in Turkey between 1990 and 2014 is shown in Fig. 2 [11]. As a result, Turkey's energy infrastructure grew and power plant capacities increased concurrently. Installed capacity of Turkey's power plants based on energy source at the end of 2015 are shown in Table 1 [12]. However, increased energy consumption parallel to Turkey's economic growth ironically had a negative effect on the country's economy. Specifically to the country's foreign trade. This is because Turkey is a net energy importer and most of the country's energy demand is supplied from imported energy sources, mostly coal and natural gas [12]. Turkey, also imports most of the oil it consumes; however, it is primarily used by the petrochemical industry for the production of transportation fuels and petrochemicals [13]. Here it should be emphasized that this problem is not unique to Turkey. Globally, three types of fossil fuels: coal, natural gas and oil provides majority of the global total primary energy supply (TPES) and the share of renewable energy sources is still small compared to these fossil fuels [14]. Turkey spent approximately 60 billion US$ for energy imports in 2013, mostly natural gas, which was

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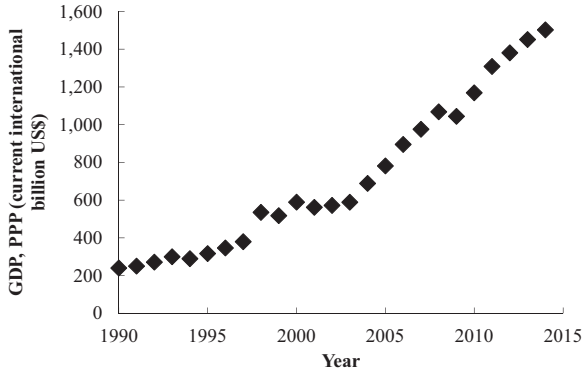


Fig. 1. Turkey's GDP (PPP) in current international billion US$ change between 1990 and 2014 [3].

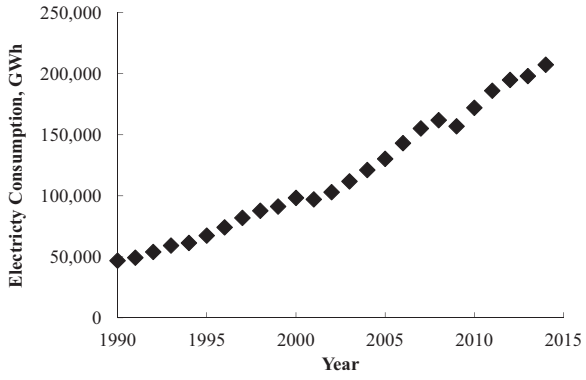


Fig. 2. Net electricity consumption in Turkey between 1990 and 2014 [11].

Table 1
Installed capacity of power plants based on energy source in Turkey at the end of 2015, MW [12].

Fuel Type	Installed Capacity, MW	Share, %	Number of Power Plants
Fuel oil + asphaltite + naphta + diesel	851.0	1.2	18
Bituminous coal + lignite	9013.4	12.3	28
Imported coal	6064.2	8.3	8
Natural gas + LNG	21,222.1	29.0	233
Renew. + waste + semi-waste + pyrolysis oil	344.7	0.5	69
Multiple fuels solid + liquid	667.1	0.9	23
Multiple fuels liquid + natural gas	3684.0	5.0	46
Geothermal	623.9	0.9	21
Hydro (dam type)	19,077.2	26.1	109
Hydro (river type)	6790.6	9.3	451
Wind	4498.4	6.1	113
Thermal (unlicensed)	56.5	0.1	24
Wind (unlicensed)	4.8	0.0	9
Solar (unlicensed)	248.8	0.3	362
Total	**73,146.7**	**100.0**	**1514**

1/4 of the import of a total of 237 billion US$ and over 75% of natural gas is being imported just from two states, Iran and Russia [15]. In the long run, this is simply not sustainable.

From energy systems engineering perspective the solution to this problem is straightforward. Turkey should decrease its imported fossil fuel consumption and increase consumption of its indigenous energy sources. However, in reality this is not an easy task. Turkey's only indigenous fossil fuel that is abundantly available is coal, mostly lignite [16]. Unfortunately, increased consumption of lignite would increase Turkey's greenhouse gas emissions. And it is known that increased consumption of fossil fuels in the 20th century has led to global climate change [17]. As a result, today, sustainability is the primary goal of global energy market [18]. This means, Turkey should decrease the share of fossil fuels and increase the share of renewable energy sources in the country's energy mix. This is also parallel to the global trend of using more renewable energy sources [19]. Globally, increasing greenhouse gas emissions, which doubled over the last three decades, and the instability of fossil fuels prices encouraged many countries to increase their investment on renewable energy sources [8].

Table 2
Turkey's Vision 2023 targets for the energy sector (target, as of 2015, can be modified) [20,21].

Items	Goals
Renewable Energy Sources	
Share of renewable sources in energy production	30%
Hydroelectric generation capacity	Maximum or 36,000 MW
Wind power installed capacity	20,000 MW
Solar power installed capacity	3000 MW
Geothermal power installed capacity	600 MW
Biomass	2000 MW
Infrastructure	
Length of transmission lines	60,717 km
Reaching a power distribution unit capacity	158,460 MVA
Use of smart grids	Established
Natural gas storage capacity	5 billion m^3
Energy stock exchange	Established
Nuclear power plants	2 operational (3rd under construction)
Installed power capacity	**120,000 MW**

In that context, the Turkish government set an ambitious target of producing 30% Turkey's energy demand in the year 2023 from renewable energy sources. This is part of Turkey's Vision 2023 energy targets, details of which are given in Table 2 [20,21]. To be specific, Turkey's 30% is significantly higher than the 20% target of the European Union (EU) set for the year 2020 [22]. Therefore, meticulous analysis on the realisation of Turkey's Vision 2023 energy targets must be carried out. So far, the author of this paper has carried out majority of this analysis on Turkey's Vision 2023 energy targets [1,16,23–26]. Here it should be emphasized that Vision 2023 targets given in Table 2 are valid as of 2015 and it is possible that some of the targets can be increased if they are achieved before 2023.

The main problem with renewable energy sources is baseload supply. Conventionally, the backbone of power systems are baseload providers, which are power plants able to generate a constant and predictable supply of electricity [27]. In most networks primary source of electrical energy are baseload plants, which use fossil fuels [28]. Therefore, increased share of renewable energy sources in Turkey's power grid could jeopardise baseload supply, if necessary precautions are not taken. As can be seen from Table 2, it is envisaged that a substantial amount of electricity will be generated from wind and solar energy to fulfil the Vision 2023 targets. To be specific, 20,000 MW from wind and 3000 MW from solar energy. Due to the intermittent nature of wind and solar energy, 7 days a week 24 h a day power supply from these renewable energy sources is problematic. Luckily, not all the renewable energy sources have such intermittency problems. Biomass, geothermal and hydropower are considered as renewable baseload energy sources [29].

Hydropower could provide baseload energy supply, conditional that there is not a continuous drought period. However, according to the Vision 2023 targets hydropower energy utilisation will be maxed out (see Table 2); and there won't be room for further improvement, if the Vision 2023 energy goals are realised. Biomass could also provide baseload energy supply conditional that enough biomass is available.

Which means, there is not a continuous drought period and there is continuous supply of biomass. And, as can be seen from Table 2 there is room for improvement there. It is suggested that nearly 2000 MW of electricity would be produced from biomass by the year 2023. Analysis of Turkey's biomass based energy supply according to Turkey's Vision 2023 energy targets prepared by the author of this paper is currently under evaluation for publication elsewhere. Therefore, in this paper that would not be analysed.

After the review on Turkey's renewable energy sources given above, we are left with geothermal energy. Luckily, geothermal energy is the only renewable energy source that is impermeable to the weather conditions. Thus it can provide a baseload energy supply [30]. However, solely the role of geothermal energy in Turkey's Vision 2023 energy targets has not yet been analysed in detail in the published literature. Turkey has a long history of electricity generation from geothermal energy. In 1984, Turkey's first and Europe's second (after Italy) geothermal power plant was opened in Denizli (Kızıldere) with 20.4 MW_e installed capacity [31]. However, its share in the country's energy mix has not reached the desired levels so far. Instead, main use of geothermal energy in Turkey is for direct use. Therefore, geothermal energy's role for power generation in Turkey must be analysed in detail. And, to do so, the development of geothermal energy around the world must be analysed in detail.

To sum up, this paper was written with a global to local perspective. First, the global development and future of geothermal energy was analysed. Then from a global to local perspective, development of geothermal energy (both direct use and power generation) in Turkey was carefully investigated. And finally, the potential role of geothermal energy in Turkey's Vision 2023 energy targets was analysed in detail.

2. Development and future of geothermal energy around the globe

In order to foster sustainable development and protect the environment, pressure is growing around the globe to speed up the utilisation of renewable energy sources [32]. In the 21st century, energy production from renewable energy sources has focused mainly on biomass, solar, wind energy and, to a lesser extent, hydropower [33]. Whereas geothermal energy has received the least attention and this may have to do with the fact that geothermal energy is available on certain locations around the globe, mostly near tectonic plate boundaries [33].

The adjective “geothermal” is derived from the Greek words; geo meaning earth, and thermal meaning heat, which can be interpreted as energy/heat produced from the earth [34]. Geothermal energy is originated from heat generated and stored in the mantle and core of the earth and deep faults let water to flow to these heat sources and the rising hot water generates a geothermal reservoir which can reach temperatures of 350 °C or higher [35]. Geothermal energy is considered as a clean, reliable, and safe renewable energy source [36,37]. Based on their reservoir fluid temperatures geothermal resources have been classified as low, medium and high enthalpy resources [38].

For a long time, utilisation of geothermal energy is classified under two groups with regard to the temperature of geothermal resources: direct use and electricity generation [39,40]. However, more recently geothermal energy systems are grouped under three main categories [41]:

i. Direct use and district heating systems: These systems use hot water from reservoirs or springs located near the surface of the earth. Historically, hot mineral springs have been used for bathing, cooking, and heating. In addition, geothermal energy is used to heat buildings through district heating systems and hot water is directly piped into buildings and industries for supplying heat. Industrially, geothermal energy is used for food dehydration, which is the most common industrial use of geothermal energy, milk pasteurizing and gold mining.
ii. Electricity generation via geothermal power plants: Electricity generation in geothermal power plants require water or steam at high temperatures (300–700 °F or approximately 150–370 °C). Therefore, geothermal power plants are usually built at the surface location of geothermal reservoirs, which are located a mile or two inside the earth's surface.
iii. Geothermal heat pumps: They utilise the constant temperatures near the surface of the earth to heat and cool buildings/structures. The principle is they transfer heat from the ground (or water) into buildings/structures during the winter and reverse the process in the summer.

Here, it should be emphasized that this paper focuses on electricity generation from geothermal energy. Specifically, understanding the global trend then assessing the past and potential applications in Turkey based on the Vision 2023 energy targets. Therefore, from the quantitative aspect we will focus mostly on that. However, in certain parts of this paper, capacity and other energy related information for direct use (heat) will be given for comparing energy values (e.g. amount of heat vs. electricity generation capacity). Because, without this information the energy analysis would be incomplete.

Non-electric or direct utilisation of geothermal energy means direct use of the heat energy from the geothermal source rather than converting it into electrical energy [42]. The major areas of direct utilisation of geothermal energy are: (i) agriculture applications, (ii) aquaculture applications, (iii) heat pumps, (iv) industrial processes, (v) space heating and cooling including district heating, (vi) swimming, bathing and balneology [43]. In late 1990s, it has been estimated installed thermal power of major direct utilisation projects exploiting geothermal energy is almost 9000 MW_t utilizing 37,000 kg/s of fluid existing in about 38 countries around the world [43]. Low, medium and high temperature geothermal sources can be used for direct utilisation [44]. For geothermal energy it was estimated that nearly 83% of the annual energy use and 84% of the installed capacity worldwide in 2005 was in district heating and China, France, Iceland, and Turkey were the leaders in district heating; whereas, Italy, Japan, Russia, and the United States were the major users in the individual home heating sector [45,46].

Historically, Prince P. G. Conti set up the first device able to produce electricity from a geothermal steam well in Larderello, Italy in the year 1904, and in the year 1914, at the same location, the first commercial geothermal power system was connected to the grid with a 250 kW turbo alternator [47]. Since then electricity generation from geothermal energy spread around the globe, and now it is more than one hundred years when electricity first generated from geothermal sources.

Geothermal power plants utilise hydrothermal resources, which have both water (hydro) and heat (thermal) [48]. The hot water or steam is used to turn a turbine and create electricity. Geothermal power plants need high temperature (300–700 °F or approximately 150–370 °C) hydrothermal resources, which come from either hot water wells or from dry steam wells [48]. Here it should be emphasized that conventional electricity generation can be achieved by using fluid temperatures above 150 °C; however considerably lower temperatures can also be used for electricity generation with the application of binary systems [44]. This will be explained below. The hydrothermal resources are accessed by drilling wells into the earth, which may be as deep as two miles, and then hot water or steam is piped out to the surface [48]. It has been reported that approximately three-quarters of prospective and productive geothermal power plants around the globe are associated with subduction zone volcanoes [49].

Today, three main technologies are in use to produce electricity from geothermal reservoirs [47,48]:

i. Dry Steam power plants: These power plants use high temperature,

vapour-dominant, hydrothermal reservoirs and steam from the well passes directly through the turbine-generator unit and generates electricity.

ii. Flash steam power plants: These power plants are used when a liquid-dominant mixture is produced at the wellhead of the hydrothermal reservoir. There are Single Flash and Double Flash power plants. In Single Flash power plants, the mixture from the wellhead is separated into different phases in a flash vessel and vapour is sent to the turbine-generator unit. The Double Flash power plants obtain more power output from the same geothermal reservoir than Single Flash power plants.

iii. Binary power plants: Binary cycle is used when the water temperature in the geothermal reservoir is lower than 150 °C. Largely binary plants are based on Kalina or Rankine cycles. In these plants, the working fluid, other than geothermal fluid, undergoes a closed cycle and it receives energy from the geothermal fluid through a heat exchanger unit and evaporates, then expands in the turbine, condenses and it is returned to the heat exchanger by a pump.

In addition to these traditional techniques, hybrid technologies with fossil fuels or other renewable energies and combined cycles are other interesting possibilities that can increase the global efficiency of geothermal power plants [47].

For more than 100 years, commercial baseload electricity is supplied from geothermal energy via conventional hydrothermal resources; and in 2010, geothermal energy had 10,751 MW_{el} of installed electricity generation capacity and 50,583 MW_{th} of direct use capacity around the globe [50]. It has been reported that, in early 2010s nearly 70 TWh of power and 0.2 EJ of heat (excluding heat pumps) are produced annually around the globe from geothermal energy [51]. Geothermal electricity net generation around the globe between 1980 and 2012 is shown in Table 3 [52].

In 2013, 70 billion kWh of electricity was generated from geothermal energy by 20 countries [41]. After the United States (U.S.), Indonesia was the 2nd largest electricity generator from geothermal energy, at nearly 11 billion kWh, which was equal to nearly 5% of Indonesia's total power generation at the same year [41]. Iceland generated nearly 5 billion kWh of electricity from geothermal energy and ranked as the 7th largest producer; however, the country had the largest share electricity generation from geothermal energy at 27% of its annual electricity generation [41]. Iceland also leads the world in geothermal development on a *per capita* basis [53].

Brief information about global geothermal energy development between 2013 and 2014 and the projection for 2020 is given as follows [54]:

- In 2013, global electricity generation from geothermal energy has surpassed 75 TWh.
- In 2014, global electricity generation from geothermal energy has surpassed 77 TWh and the total cumulative capacity exceeded 12 GW.
- In 2014, newly installed geothermal energy capacity was nearly 0.8 GW in 2014.
- It is estimated that global geothermal electricity capacity will rise to over 16 GW in 2020 and provide nearly 104 TWh globally.

Today, the U.S. is the world leader for geothermal energy based electricity generation; in 2015, nearly 16.8 billion kWh, or 0.4% of total U.S. electricity generation is supplied from geothermal energy [41]. In 2015, seven states of the U.S. had geothermal power plants and share of geothermal electricity produced by each state were roughly: California 74%, Nevada 20%, Utah 3%, Hawaii 1.4%, Oregon 1.1%, Idaho 0.6%, New Mexico 0.1% (does not add up to 100% due to rounding) [41].

Detailed analysis of the literature showed that there is a huge potential for development and utilisation of geothermal energy around the globe. Increasing environmental consciousness and concern about sustainability indicates that the share of renewable energy sources and concurrently geothermal energy will increase in the global energy mix of tomorrow. It has been estimated that geothermal energy can account for nearly 3.5% of global annual electricity production and 3.9% of energy for heat (excluding ground source heat pumps) by the year 2050, which would be a significant increase from 2011 levels of 0.3% and 0.2%, respectively [55]. Also, International Energy Agency (IEA) estimated that electricity generation from geothermal energy could increase to 1400 TWh per annum in 2050, which would be equal to 3.5% of global annual electricity generation and avoid nearly 800 megatonnes (Mt) of CO_2 emissions per annum [56]. IEA also estimated that that by 2050 geothermal heat could contribute 5.8 EJ (1600 TWh thermal energy) per annum, which would be nearly 3.9% of projected final energy for heat [56].

However, geothermal energy is not a miracle solution. Utilisation of geothermal energy may not always be continuous, because its sustainability depends long term replenishment rates and to the sustainability of site specific heat extraction [35,57]. This means excess use (unsustainable rates of utilisation) of the geothermal source for electricity generation may temporarily deplete the energy source and the geothermal resource cannot provide sustainable energy supply [57].

Consequently, sustainability of geothermal energy is a major concern and in a recent study the following points are highlighted about this subject [58]:

- Large scale geothermal plants can cause serious environmental problems such as discarding of the geothermal brines or deposition of salts and silica.
- There are uncertainties about the supposed positive effect of geothermal energy on climate change and related mitigating policies.
- There are studies showing that traditional, dry-air or flash steam geothermal plants are not carbon neutral because of both embodied and fugitive greenhouse gas emissions.

Table 3
Geothermal electricity net generation (billion kilowatt hours) around the globe between 1980 and 2012 [52].

	Geothermal Electricity Net Generation (Billion kilowatt hours)							
Years	**1980**	**1985**	**1990**	**1995**	**2000**	**2005**	**2010**	**2012**
Region								
North America	5.9	11.0	20.6	19.0	20.0	22.0	21.8	21.4
Central & South America	0.5	0.7	0.8	1.3	2.1	2.7	3.4	3.8
Europe	2.6	2.7	3.6	3.9	6.2	7.1	10.7	11.9
Middle East	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Africa	0.0	0.3	0.3	0.4	0.4	1.0	1.5	1.6
Asia & Oceania	4.2	7.6	10.5	13.7	22.8	23.3	28.4	29.0
World	**13.2**	**22.3**	**35.8**	**38.3**	**51.6**	**56.6**	**66.3**	**68.2**

In order to sustainably operate geothermal power plants geothermal fluids should be re-injected into the reservoirs. Also, it is known that geothermal fluids have physicochemical properties, which are not compatible to the terrestrial ecosystems; therefore, after utilisation for power generation geothermal fluids should be treated and re-injected into the geothermal reservoirs [59]. However, these processes have high economic costs because they require drilling and maintenance of additional wells, plus treatment and pumping of geothermal fluids [59].

Reinjection is crucial for all types geothermal reservoirs under production and it is vital in the success or failure of the geothermal power development [60]. Reinjection of water into a geothermal system during operation is aimed to serve two purposes: (i) waste water disposal and (ii) improved resource recovery, and this can be accomplished via good field management and a thorough design of the reinjection system [60].

In order to assist with the evaluation of reinjection effects, geothermal reservoirs are classified under 3 categories by Kaya and co-workers [61] in terms of temperature (T) and production enthalpy (h), which is essentially the specific enthalpy because its unit is kJ/kg. According to their classification the first category is hot-water ($T < 220$ °C and $h < 943$ kJ/kg), the second category is two-phase, liquid-dominated, which has 3 subcategories, low-enthalpy (220 °C $< T <$ 250 °C and 943 kJ/kg $< h <$ 1100 kJ/kg), medium-enthalpy (250 °C $< T <$ 300 °C and 1100 kJ/kg $< h <$ 1500 kJ/kg), and high-enthalpy (250 °C $< T <$ 330 °C and 1500 kJ/kg $< h <$ 2600 kJ/kg), and the third category is two-phase, vapour-dominated (250 °C $< T <$ 330 °C and 2600 kJ/kg $< h <$ 2800 kJ/kg) [61].

If the reinjection is carried out properly and the geothermal fields are utilised carefully then geothermal energy can be used continuously for a long period. Thus helping us move to a more sustainable future by partially supplying our increasing global energy demand from a renewable baseload energy source.

In this section, the development and future of geothermal energy around the globe is analysed and the advantages and disadvantages of geothermal energy development is investigated. Therefore, the development and future of geothermal energy in Turkey by using the valuable information given in this section and scrutinise its role in Turkey's Vision 2023 energy targets can be analysed now.

3. Development and future of geothermal energy in Turkey based on Vision 2023 energy targets

In Turkey, exploration of geothermal energy started in early 1960s [62]. At the beginning work was focused on high enthalpy fields for potential power production and Kızıldere field was discovered in 1968, then Balçova and Seferihisar, two medium temperature fields, were located and analysed in 1960s and 1970s, and until today more than 200 geothermal fields were discovered [62,63]. Geothermal systems of Turkey primarily follow recent and regional structural lines and they are more “frequent in regions of recent tectonism and Tertiary volcanism and/or metamorphism” [62].

General information about Turkey's geothermal energy potential is as follows [64,65]:

- Turkey is the 7th richest country in terms of geothermal potential around the world.
- Work on geothermal energy started by the General Directorate of Mineral Research and Exploration (MTA) in 1962.
- It has been reported that 95% of the geothermal fields that have been discovered by MTA are low-medium enthalpy fields that are mostly suitable for direct use applications.
- So far, 190 geothermal fields have been located in Turkey by the MTA.79% of these fields are located in Western Anatolia region, 8.5% in Central Anatolia region, 7.5% in Marmara region, 4.5% in Eastern Anatolia region and 0.5% in other regions.
- 94% of Turkey's geothermal resources are low and medium temperature, which are suitable for direct use (heating, thermal tourism, mineral extraction) and the remaining 6% is suitable for power generation.
- So far, drilling work in the geothermal fields yielded in the production of 3881 MW_t of thermal energy, which is equivalent to nearly 12.3% of Turkey's geothermal thermal energy potential. 30% of this thermal energy is used for heating houses and thermal spas in locations like İzmir, Gönen, Simav, Kırşehir, Kızılcahamam, Afyon merkez, Sandıklı, Kozaklı, Diyadin, Salihli, Edremit, Sarayköy, Bigadiç(nearly 100,000 house equivalent), greenhouses (nearly 1,000,000 m^2) and health and thermal tourism (215 establishments).
- Turkey's overall geothermal potential is estimated between 31,500 MW [65] and 38,000 MW [64].
- Nearly, 88% of this potential is suitable for thermal use, temperature less than 473 K, and the remainder is suitable for electricity generation, temperature more than 473 K.
- Turkey's geothermal energy potential suitable for power production is estimated at nearly 4500 MW.

Geothermal electricity net generation in Turkey between 1980 and 2012 is given in Fig. 3 [52]. As can be seen from Fig. 3, geothermal energy production in Turkey increased almost exponentially starting from 2006. Percentage share of geothermal energy in Turkey's electricity net generation between 1980 and 2012 is calculated and given in Table 4 [52,66]. As can be seen from Table 4 percentage share of geothermal energy in Turkey's electricity net generation increased from nearly 0.05% in 2006 to nearly 0.38% in 2012, which means nearly 8 folds increase in 6 years.

After many coalition governments, Turkey's Justice and Development Party (AKP) rise to power in 2002. Since then Turkey has been governed by a single party. During the rule of AKP Turkey's economy grew at record levels and the country's energy consumption increased concurrently. As can be seen from Fig. 2, in 2002, electricity consumption in Turkey was nearly 103,000 GWh and this rose to nearly 207,000 GWh in 2014 [11]. As can be seen from Table 1 and Fig. 3, the share of renewable energy sources and specifically geothermal energy (as the primary concern in this paper) increased concurrently. To summarise, the following changes happened in Turkey's geothermal energy applications between 2002 and 2015 [67]:

- Number of fields suitable for electricity production increased from 16 in the year 2002 to 25 in the year 2015.
- In 2002, nearly 500,000 m^2 of greenhouse space is heated via geothermal energy. In 2015, this increased to 3,931,000 m^2.
- Residential heating from geothermal energy increase from 30,000 residence equivalent in 2002 to 114,567 residence equivalent in

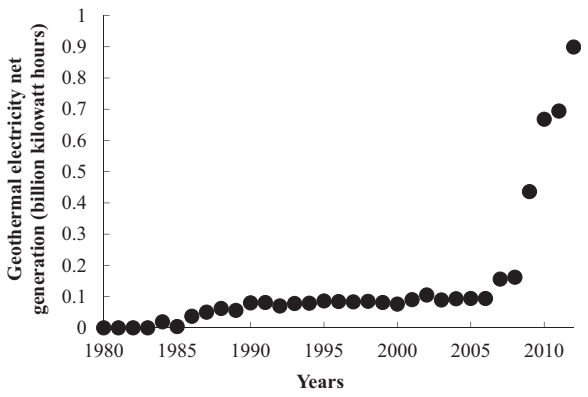


Fig. 3. Geothermal electricity net generation (billion kilowatt hours) in Turkey between 1980 and 2012 [52].

Table 4
% share of geothermal energy in Turkey's electricity net generation between 1980 and 2012 [52,66].

Years	% of Turkey's Electricity Net Generation
1980	0.000
1981	0.000
1982	0.000
1983	0.000
1984	0.062
1985	0.012
1986	0.093
1987	0.113
1988	0.129
1989	0.108
1990	0.139
1991	0.134
1992	0.104
1993	0.106
1994	0.101
1995	0.100
1996	0.089
1997	0.080
1998	0.077
1999	0.070
2000	0.061
2001	0.073
2002	0.081
2003	0.063
2004	0.062
2005	0.058
2006	0.053
2007	0.081
2008	0.082
2009	0.224
2010	0.316
2011	0.303
2012	0.375

2015.

- Electricity production from geothermal energy increased from 15 MW_e in 2002–612.8 MW_e in early 2015.
- Geothermal heat capacity increased from 3000 MW_t in 2002 to 14,000 MW_t in 2015.

This growth in the geothermal energy utilisation is related to the economic growth but also to the new legislations. A significant development in geothermal electricity production and direct uses (district, greenhouse heating and thermal tourism) was achieved in Turkey during the last couple of years, particularly after the geothermal law issued in 2007 and its regulations, feed in tariff policies [68–70]. Nearly, 3,200 licences for Turkey's geothermal field were given to the private sector after the law become effective by the MTA, which generated nearly 600 million U.S. dollars of income for the government [71].

Another law was passed in 2010, Law No: 6094, Date of Acceptance: 29/12/2010, Official Gazette Publishing Date: 08/01/2011, No: 27809, which issued tariff subsidies as well as purchasing guarantees for electricity generation via renewable energy resources [72]: "The government announced a 10.5 ¢/kWh subsidized tariff and 10 year purchasing guarantee from the date the plants are commissioned for power generation via geothermal resources until 2020" [71]. There is also additional local product support suitable for geothermal energy equipment manufactured in Turkey, which is also valid for 10 years: steam or gas turbine 1.3 ¢/kWh, generator and power electronics 0.7 ¢/kWh, steam injector or vacuum compressor 0.7 ¢/kWh [71,72]. Thus totalling the subsided tariff to 13.2 ¢/kWh [71,72]. The purchasing guarantee and tariff subsidies have certainly improved project financing for local and foreign investors interested in geothermal energy development in Turkey [71,72]. With the help of these laws and other renewable energy laws passed so far geothermal energy market of Turkey grew at record levels in the last 10 years.

However, as can be seen from Table 2, geothermal energy has the smallest share in renewable energy sources (in terms of installed capacity) in the Vision 2023 energy targets. This is mostly related to the limited potential capacity of this renewable energy source for electricity generation, nearly 4500 MW as stated above. Considering that Turkey's installed capacity would rise to 120,000 MW and the planned geothermal energy installed capacity would be 600 MW (Table 2) in 2023 (target, as of 2015, can be modified), geothermal power plants could only provide 0.5% of Turkey's installed capacity. This means geothermal energy could not provide a nationwide supply; however, it is evident that it would provide local benefits and it is crucial for Turkey to develop most of this potential.

As can be seen from Table 1, Turkish Electricity Transmission Corporation (TEIAS) reported that Turkey's 21 geothermal power plants total installed capacity of reached to 623.9 MW supplying nearly 0.9% of Turkey's total installed capacity of 73,147.6 MW at the end of 2015 [12]. As can be seen from Table 2, Turkey's Vision 2023 target for geothermal installed capacity is 600 MW (target, as of 2015, can be modified) and at the end of 2015 Turkey has successfully surpassed that target. This is important for the realisation of the Vision 2023 energy targets; and critically, this is the first renewable energy target that has been fulfilled before the year 2023.

A detailed short analysis about the role of geothermal energy in Turkey's Vision 2023 energy targets is given in a recent publication by the author of this paper [25]. And in that publication the author of this paper has pinpointed that the geothermal electricity generation potential of those who had obtained licences from the Energy Market Regulatory Authority (EMRA) is expected to increase to 1000 MW by the end of 2023 [25]. Therefore, in this study, the author is suggesting to modify the Vision 2023 targets and increase the goal for the geothermal installed capacity to increase to 1000 MW. By this way, the share of geothermal energy in Turkey's installed capacity of the year 2023 could rise to nearly 0.8%. Also, the motivation of private and government sector in the realisation of the Vision 2023 targets can be strengthened.

4. Conclusion

After a global to local analysis the following conclusions were drawn from this study. It was targeted that geothermal energy installed capacity would reach 600 MW by the year 2023 according to the Vision 2023 energy targets (target, as of 2015, can be modified). In this study, it was pinpointed that geothermal energy installed capacity of Turkey reached to 623.9 MW at the end of the year 2015. This is significantly important for the realisation of the Vision 2023 energy targets, and it is the first renewable energy target that has been accomplished before the year 2023. Moreover, in this study, it was suggested that geothermal energy target for the Vision 2023 agenda can be modified to increase the installed capacity aim to 1000 MW for the year 2023. By this way, the share of geothermal energy in Turkey's installed capacity of the year 2023 could rise to nearly 0.8% in 2023.

Consequently, it can be stated that there are both opportunities and problems ahead of geothermal energy development around the globe and locally in Turkey. However, when the obstacles are overcome it is evident that geothermal energy could help us move to a more sustainable future by providing the energy that we need to drive our economies.

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I would like to dedicate this paper to my lovely wife Bengi Melikoglu. I wish you health.

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